

AURA Forge

Governed Adaptive Engineering for FitGPT

LaunchCode Agentic Engineering Capstone
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**Give each
software
change only
the autonomy
it deserves.**

Fixed autonomy made simple changes too

The same route was being applied to changes with very different risk.

expensive



LOW

Full route wasted time and cost on a documentation-only change.

HIGH

Removing governance would have been unsafe for policy-sensitive work.

AURA Forge routes each change by risk

A deterministic classifier selects risk before any route is chosen.

AURA Forge architecture



Evidence boundary

Every route emits structured evidence.

Policy, eval, test, approval, and CI signals feed the Change Passport.

No unsupported production claims.

Core idea

The classifier does not implement code, grant tools, or call a model. It only determines LOW, MEDIUM, or HIGH. The router then assigns roles, tools, gates, and human checkpoints.

LOW gets speed; HIGH keeps governance

The route is lighter only when the risk is lower.

Adaptive route matrix

The route changes only when risk justifies it.

LOW	Implementer -> Reviewer -> deterministic gates
Human checkpoints	0
MEDIUM	Implementer -> Reviewer -> Tester -> final approval
Human checkpoints	1
HIGH	Planner -> approval -> Implementer -> Reviewer -> Tester -> gates
Human checkpoints	2

HIGH is not optimized away.

The experiment compared matched scenarios

Control first, freeze the router, then measure AURA.

AF-LOW-001

Documentation /
accessibility

AF-MEDIUM-00

1
Bounded
executable test
change

AF-HIGH-001

Governance / MCP
policy-sensitive

Matched rules

Same locked 16-point quality rubric. Control route measured first. Router frozen before AURA measurements. Infrastructure failures preserved separately.

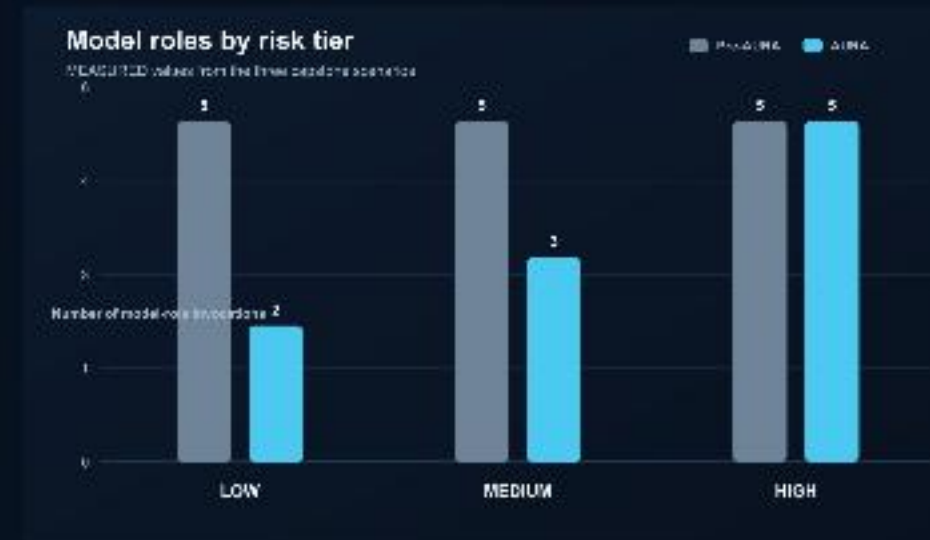
Measured results: less overhead, stronger quality

Measured impact across the three representative scenarios.



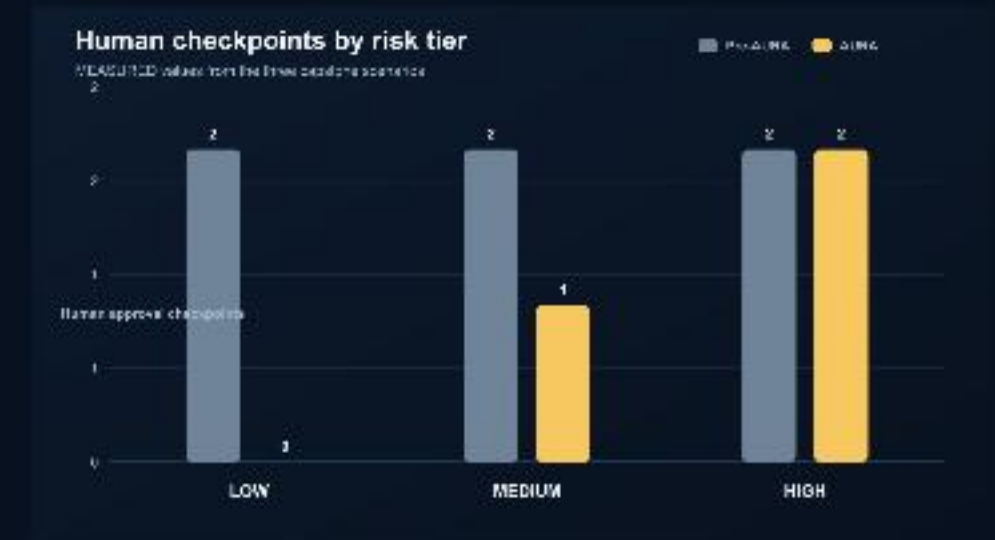
19.22%

cost reduction



33.33%

fewer model roles



50%

fewer human
checkpoints

44/48 -> 48/48

quality improved

Governance stopped a real overreach attempt

The runtime denied a Project-Manager-only action from an implementer.

Governance denial evidence

GO-20260811-001 used the real governed authorization layer.

Role	implementer
Attempted tool	task_tracker
Decision	DENIED
Reason	only project-manager is allow-listed
External state changed	No
Model cost	\$0

Sanitized denial

Authorization error: role 'implementer' is not on the allow-list for task_tracker.
Allowed roles: ['project-manager'].

This matters because the role that changes files cannot also mark the work complete.

CI made the engineering gates visible

The final branch passed the permanent governance workflow.

GitHub CI summary

Final engineering CI evidence.

Workflow	AURA Forge Governance CI
Run	31517900793
change-classifier	PASS
policy-tests	PASS
evaluation-gate	PASS
pipeline-integrity	PASS
audit-trail	PASS

CI path

**first failed CI ->
diagnosis -> repair ->
green CI**

Final run 31517900793 passed change classification, policy tests, evaluation, pipeline integrity, audit trail, and advisory graceful skip.

Change Passport proves readiness

Readiness is backed by machine evidence, not a claim.

Change Passport summary

Tracked sanitized example: AF-HIGH-001.

Scenario	AF-HIGH-001
Readiness	PASS
Quality	16/16
Risk tier	HIGH
Route	aura-high-v1
Policy tests	17 passed
Human checkpoints	2

Evidence flow

AURA Forge treats proof as an output, not an afterthought.

Classifier	risk tier + triggered rules
Router	route id + roles + checkpoints
Gates	policy, eval, tests, holdout checksum
Approvals	human events only where required
Passport	readiness summary with evidence hashes

A separate deterministic conversion showed one narrow factual check could run with \$0 model cost and about 0.156 ms local runtime.

Impact is measured, bounded, and honest

The best agentic system knows when not to use agents.

Measured impact

Quality improved. LOW and MEDIUM overhead dropped. HIGH governance stayed in place.

Limitations

Three representative scenarios only. No production deployment. Advisory AI was unavailable in CI and intentionally degraded safely.

Final takeaway: AURA Forge gives each change the minimum autonomy it deserves, then proves the route stayed inside policy.